

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 1.1 REAL NUMBERS: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why can we never write some everyday numbers down completely, and how do mathematicians and Kenyan professionals work accurately with numbers that go on forever? Use what you learned in Lessons 1 to 8 to answer the five sections below. Refer to the Ugali Pot Problem from Gikomba Market: a sufuria with a diameter of 28 cm, where $28 \times \pi = 87.96459430\dots$ on the calculator and the daughter cut her string to 88 cm. Write in full sentences, show every step of any calculation, and use the correct mathematical words. Use your own words and the evidence from our class work.

Part 1: Which numbers can be written down exactly?

Look at the table of everyday Kenyan numbers.

Number	Where it comes from
60.50	Price of 1 kg of unga (Ksh)
160	Distance from Nairobi to Nakuru (km)
34.5	Temperature in Kisumu (°C)
1/3	Share of a kilo of unga
$\sqrt{2}$	Diagonal of a square with sides of 1 m

(a) Rational: 60.50, 160, 34.5 and $1/3$. Irrational: $\sqrt{2}$ and π .

(b) Rational numbers have decimals that either terminate (60.50, 34.5, 160) or repeat in a pattern ($1/3 = 0.333\dots$). Irrational numbers have decimals that never terminate and never repeat ($\pi = 3.14159\dots$, $\sqrt{2} = 1.41421\dots$).

(c) A number is rational if it can be written as a fraction p/q where p and q are integers and $q \neq 0$. Irrational numbers cannot be written in this form. Both kinds belong to the real numbers, so every number in the table is a real number.

π	Used for the circumference of the sufuria	
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(a) Classify each number as rational or irrational. (b) Describe what the decimal form of each type of number looks like. (c) State the test that decides whether a number is rational.

Part 2: Why the sufuria circumference cannot be written completely

The sufuria has a diameter of exactly 28 cm, which is a nice whole number. Explain why its circumference cannot be written as an exact fraction or a terminating decimal. In your answer, name the type of each number in the calculation $28 \times \pi$ and say what kind of number the product is. Also compare with $28 \times 3 = 84$.	The circumference is $28 \times \pi$ cm. The number 28 is rational (a whole number, so it can be written as $28/1$). π is irrational: it cannot be written as p/q, and its decimal never ends or repeats. A rational number multiplied by an irrational number is generally irrational, so 28π is irrational. This is why the calculator shows 87.96459430... with digits that go on forever without a pattern, and no fraction or terminating decimal can equal it exactly. The product $28 \times 3 = 84$ is exact only because 3 is a rational number that is not π. It is just a rough replacement for π, not the real value. So a whole-number diameter does not guarantee a 'nice' circumference. What matters is whether the other number in the product is rational or irrational.
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Part 3: Working accurately with numbers that never end

The daughter in Gikomba rounded her answer to 88 cm. (a) Explain what she changed about the type of number when she did this. (b) Show what happens if she uses $22/7$ in place of π, and say what type of number you get. (c) Explain why a civil engineer in Kisumu sizing a water-tank rim should be careful about when to round,	(a) 28π is irrational and cannot be written completely. By rounding to 88 she replaced it with a rational number that can be written exactly. We gain a number we can write and use, but it is only an approximation, so a small error is introduced. (b) $28 \times 22/7 = 4 \times 22 = 88$. This is a rational number, so it can be written exactly, but $22/7 = 3.142857...$ is only close to π and is not equal to it. The answer 88 is therefore close to the true circumference, not exact. (c) BODMAS works the same way for irrational numbers as for rational ones (the same rules that handle Ksh 60.50×2.5 handle $\pi \times 28$). Rounding π early, for example to 3.142, and then continuing the calculation makes the error grow through later steps. The most accurate method is to keep the exact form (such as 28π) or the calculator's full decimals, and round only at the final answer, to the accuracy that the job needs.
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using what you learned about BODMAS and rounding.	
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Part 4: Measuring how wrong the approximation is	
Use $28\pi = 87.96459430\dots$ to find the error in the daughter's 88 cm string. (a) Write the expression for the error using absolute value, and calculate it to 3 decimal places. (b) Is the error a rational or an irrational number? Explain using what you learned about combining rational and irrational numbers. (c) Explain why absolute value is a suitable tool for stating the size of an error.	(a) Error = $88 - 28\pi = 88 - 87.96459430\dots = 0.03540570\dots \approx 0.035$ cm. The value inside is positive, so $x = x$ and the string is about 0.035 cm longer than the exact circumference, giving a tiny overlap. (b) The error is irrational. It is 88 (rational) minus 28π (irrational), and a rational minus an irrational is always irrational. So its decimal also never ends or repeats, and the error cannot be written as an exact fraction either. (c) Absolute value is the distance of a number from zero on the number line, defined as $x = x$ if $x \geq 0$ and $x = -x$ if $x < 0$, so it is always non-negative. The absolute difference $a - b$ gives the distance between a working value and the true value, whether the working value is too big or too small. It does not change whether a number is rational or irrational.

Part 5: Final explanation of the driving question	
Write a clear final explanation that answers the driving question: Why can we never write some everyday numbers down completely, and how do mathematicians and Kenyan professionals work accurately with numbers that go on forever? Use at least four ideas from the lessons, including the sufuria in Gikomba Market, the real number system (rational and irrational numbers), the number line, and the use of absolute value. Mention at least one Kenyan professional.	The real number system contains rational numbers and irrational numbers. Rational numbers, such as Ksh 60.50 for unga, 160 km from Nairobi to Nakuru and 34.5°C in Kisumu, can be written as p/q and have terminating or repeating decimals, so we can write them down exactly. Irrational numbers such as π and $\sqrt{2}$ cannot be written as p/q, and their decimals never end or repeat. The sufuria in Gikomba has a diameter of 28 cm, which is rational, but its circumference 28π is a rational number multiplied by an irrational number, so it is irrational. That is why the calculator shows 87.96459430... forever and why the daughter's string always has a tiny gap or overlap. Every irrational number still has one exact point on the number line, but we can only locate it to a chosen accuracy by using decimal approximations, so we can get closer but never write it completely. To work accurately, professionals keep the exact form (such as 28π) or full decimals through a calculation following BODMAS, round only at the end, and then use absolute value to state the error, for example $88 - 28\pi \approx 0.035$ cm. An engineer in Thika or Kisumu, a pharmacist in Kisumu or a trader in Eldoret can then say exactly how close their working value is to the true value. So we cannot write some numbers completely, but we can always control and measure how much we are off.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Classifying real numbers	Correctly classifies all numbers as	Classifies most numbers correctly and	Classifies some numbers correctly but

	rational or irrational, with the p/q test and decimal behaviour (terminating, repeating, non-terminating non-repeating) explained accurately.	mentions either the p/q test or the decimal behaviour.	gives little or no reason, or confuses recurring decimals with irrational numbers.
Explaining the phenomenon	Clearly explains that 28π is irrational because it is a rational multiplied by the irrational π , and links this to the endless decimal and the gap or overlap in the string.	States that π is irrational and links it to the endless digits, with a partial explanation of the product.	Describes the calculator result but does not link it to irrationality.
Accuracy and approximation	Explains the effect of rounding or using $22/7$, shows $28 \times 22/7 = 88$, and gives the strategy of carrying exact values and rounding only at the end.	Explains the approximation idea and gets most calculations correct, but the rounding strategy is not fully explained.	Calculations are incomplete or incorrect, and rounding is not explained.
Absolute value and error	Correctly calculates $ 88 - 28\pi \approx 0.035$ cm, states it is irrational with a reason, and explains why absolute value suits error measurement.	Calculates the error correctly but gives an incomplete reason for its type or for the use of absolute value.	Does not use absolute value correctly, or the error calculation is incorrect.
Final explanation and use of evidence	Gives a coherent explanation of the driving question using at least four lesson ideas and Kenyan contexts, with accurate mathematical vocabulary.	Gives a mostly clear explanation using two or three ideas from the lessons.	Gives a general or incomplete answer with few ideas from the lessons or little use of Kenyan contexts.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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